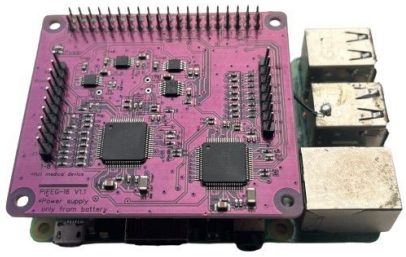
PRODUCT DATASHEET · SKU CQA240229G16

16-Channel EEG Shield for Raspberry Pi

Open-source Raspberry Pi shield that turns a Raspberry Pi 4 or 5 into a 16-channel brain-computer interface. Two Texas Instruments ADS1299 converters read EEG, EMG and ECG at 24 bits across the whole head, and the Pi streams, records and processes the data in real time.



16	2 × ADS1299	250 SPS	SPI0	Default
Channels + REF + BIAS	24-bit, 8 channels each	Default, up to 16 kSPS	40-pin GPIO header	PiEEG Server device

01 OVERVIEW

Key features

- **Full 10-20 coverage**: 16 channels across front, centre, temples and back of the head, plus reference and bias.
- **Two ADS1299 converters**: 2 × 8 simultaneous 24-bit channels, programmable gain 1–24.
- **Plugs straight onto the Pi**: Raspberry Pi 4 or 5, no cables between shield and host.
- **Default PiEEG Server device**: one command for dashboard, WebSocket, CSV, LSL and OSC.
- **Raw access**: unfiltered data and full register control from the Python SDK.
- **Wearable option**: printable Pi 5 case that mounts on the open-source IronBCI headset.

Typical applications

- Mapping activity across the head
- BCI control: robot arm, drone, exoskeleton, toy car
- Alpha, blink and motor experiments
- Teaching neuroscience and signal processing
- EMG and ECG projects
- Machine learning on multichannel EEG

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	16 single-ended biosignal inputs in 2 groups of 8
Extra inputs	Reference (SRB1, common to all channels) and bias drive
Analog front end	2 × Texas Instruments ADS1299, 8 channels each, independent (not daisy-chained)
Resolution	24-bit delta-sigma
Sample rate	250 SPS set by PiEEG software (ADS1299 supports 250 SPS to 16 kSPS)
Gain	1 set by PiEEG software (ADS1299 supports 1, 2, 4, 6, 8, 12, 24)
Reference voltage	4.5 V internal; 1 LSB ≈ 0.268 µV at gain 1
Signals	EEG, EMG, ECG
HOST INTERFACE	
Host	Raspberry Pi 4 or Raspberry Pi 5
Connector	40-pin GPIO header (2 × 20, 2.54 mm), stacking
Bus	SPI0 shared by both chips · mode 1 · CS: CE0 and GPIO 19 · DRDY: GPIO 26 and GPIO 13

For research and education only. Not a medical device. Specifications subject to change.

Data frames	2 × 27 bytes per sample (one per chip): 3-byte status + 8 × 3-byte channels
ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connectors	Left header: BIAS, REF, channels 1–8 · right header: channels 9–16 · 2.54 mm
POWER & SAFETY	
Supply	5 V through the Raspberry Pi, from a 5 V / 3 A power bank only
Indicators	Status LEDs, on when powered
Isolation	None. Battery operation of the whole setup is required
Classification	Research and education tool. Not a medical device
MECHANICAL	
PCB	Raspberry Pi HAT footprint, nominal 65 × 56.5 mm
Mounting	4 holes on the 58 × 49 mm Pi pattern (M2.5)
Enclosure	Printable Pi 5 case with headset-mount lid (see PE16-ENC-007)
Weight	100 g (Elecrow listing)
SOFTWARE	
PiEEG Server	`pieeg-server` (PiEEG-16 is the default) — dashboard :1617, WebSocket :1616, CSV, LSL, OSC
Python SDK	pieeg16_drdy_only.py, Save_data_OS.py, command-line CSV recorder
BrainFlow	Not supported natively (PIEEG_BOARD is the 8-channel board); use LSL from PiEEG Server

03 BLOCK DIAGRAM

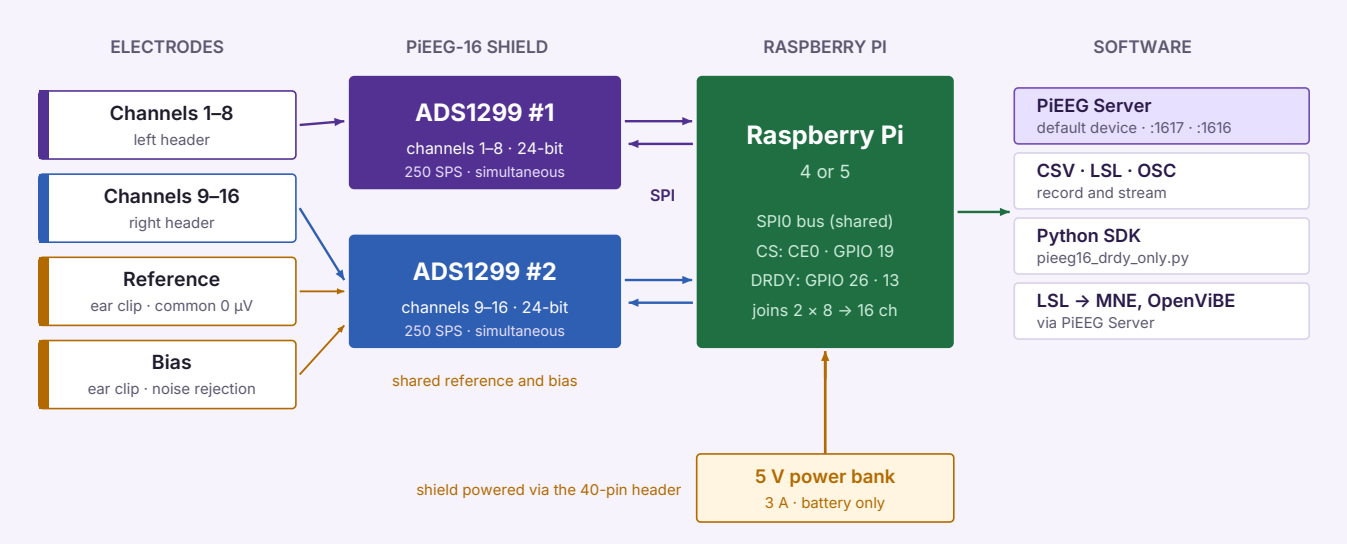


Figure 1 — Two ADS1299 chips share the SPI bus, each with its own chip select and data-ready line. The Pi joins both halves into one 16-channel frame.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms across the whole head	Close the eyes: alpha (8–13 Hz) rises on O1, O2, P3, P4
EMG	Muscle activity	Clench the jaw — strongest on T7, T8
ECG	Heart activity	Electrodes on the chest or wrists

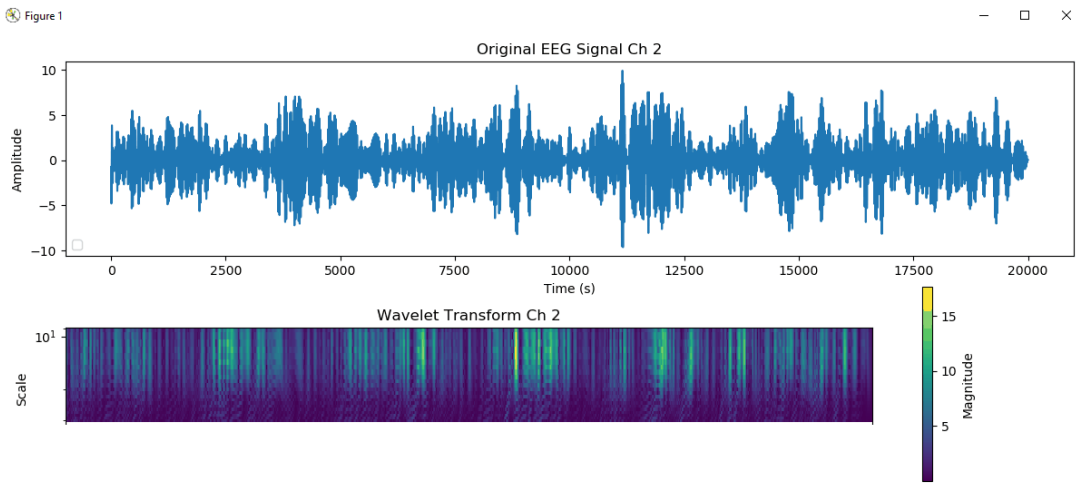


Figure 2 — Alpha test with dry electrodes: the bright bands in the wavelet view are the eyes-closed periods.

05 PIEEG FAMILY

	PiEEG-8	PiEEG-16	IronBCI-32
Channels	8	**16**	32
Host	Raspberry Pi	**Raspberry Pi**	Laptop (USB)
Converters	1 × ADS1299	**2 × ADS1299**	4 × AD7771
Sample rate	250 SPS	**250 SPS**	~500 SPS
BrainFlow	Yes (ID 56)	**No — use LSL**	Yes (ID 65)
List price	USD 350	**USD 390**	USD 480

06 ORDERING INFORMATION

Product	PiEEG-16 — 16-channel EEG shield for Raspberry Pi
SKU	CQA240229G16
Price	USD 390 (pieeg.com list price)
Purchase	pieeg.com/store/pieeg-16 · Elecrow (official manufacturing partner) — elecrow.com/pieeg-16.html
Accessories	16-channel EEG cap kit (electrodes and ear clips included)

Included	
1 × PiEEG-16 shield	2 × ADS1299, 40-pin header, 2 electrode headers

Not included — required	
Raspberry Pi 4 or 5	microSD with Raspberry Pi OS
5 V / 3 A power bank	Battery only
Electrodes and cap	16 Ag/AgCl + 2 ear clips
Monitor, keyboard	Battery-powered, wired

WARNING Battery only.

PiEEG-16 has no isolation. The Raspberry Pi, the shield and everything cabled to them must run from a 5 V power bank while electrodes are on a person. PiEEG-16 is not a medical device.

07 RELATED DOCUMENTS

PE16-MD-002 Mechanical & Dimensional Drawing	**PE16-PCB-003** PCB & Manufacturing Specification	**PE16-PIN-004** Connector Pinout & Electrode Guide
PE16-QSG-005 Quick Start & Safety Guide	**PE16-SW-006** Data Interface & Software Guide	**PE16-ENC-007** 3D-Printed Enclosure Guide